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$$\begin{aligned}& \int x^2 e^x dx \\& \left\{ \begin{array}{l} u = x^2 \\ dv = e^x dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = 2x dx \\ v = e^x \end{array} \right. \\& = x^2 e^x - 2 \int x e^x dx \\& \left\{ \begin{array}{l} u = x \\ dv = e^x dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = dx \\ v = e^x \end{array} \right. \\& = x^2 e^x - 2 \left(x e^x - \int e^x dx \right) \\& = x^2 e^x - 2x e^x + 2e^x + C\end{aligned}$$

References